

Seth M. Bushinsky

Curriculum vitae

CONTACT INFORMATION

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EDUCATION

2015	Ph. D., Oceanography, University of Washington, Seattle, WA
2011	M. S., Oceanography, University of Washington, Seattle, WA
2006	B. S., Biological Sciences, honors in Marine Biology, Stanford University, Stanford, CA

PROFESSIONAL HISTORY

2025-present	Associate Professor, Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, HI
2026	Visiting Professor, Department of Environmental Systems Science, Eidgenössische Technische Hochschule (ETH) Zürich, Zürich, Switzerland
2019-2025	Assistant Professor, Department of Oceanography, University of Hawai'i at Mānoa, Honolulu, HI
2024-2025	Affiliate Graduate Faculty, School of Earth and Ocean Sciences, University of Victoria, Canada
2018-2019	Associate Research Scholar, Program in Atmospheric and Oceanic Sciences, Princeton University, Princeton, NJ
2015-2018	Postdoctoral Research Associate with Dr. Jorge Sarmiento, Program in Atmospheric and Oceanic Sciences, Princeton University, Princeton, NJ
2006-2008	Research Technician for Dr. Francisco Chavez, Biological Oceanography Group, Monterey Bay Aquarium Res. Institute, Moss Landing, CA

HONORS AND AWARDS

2025	AMS Scientific and Technical Activities Committee (STAC) Outstanding Early Career Scientist award from the Committee on Air-Sea Interaction: <i>For improving understanding and measurement capabilities of air-sea gas fluxes and their impact on ocean biogeochemistry.</i>
2024	National Academy of Sciences Kavli Fellow.

RESEARCH OUTPUT

Student advisees^{*}, Postdoc advisees[#]

Manuscripts submitted / in review

6. König[#], D., Jutras[#], M., **Bushinsky, S. M.**, Cerovečki, I. Accumulation pre-subduction drives high anthropogenic carbon content in Subantarctic Mode Water. In review at *Global Biogeochemical Cycles*.
5. **Bushinsky, S. M.**, Z. Nachod^{*}, M. Jutras[#], D. König[#], S. McClish^{*}, and C. Addey^{*}. BGC-Argo+: Global biogeochemical Argo float data with secondary quality control and derived parameters. In review at *Earth System Science Data*.
4. Le Chevere et al. Exploiting depth-resolved observations from Argo floats and ships for a fresh approach to model evaluation of the ocean carbon cycle. In Review at *JAMES*.
3. Jutras[#], M., **S. M. Bushinsky**, I. Cerovečki, Z. Nachod^{*}, and N. Hawco. Widespread increase of nitrate in subsurface waters over recent decades revealed by observations. In review at *Nature Geoscience*. DOI: 10.22541/essoar.177100488.81138870/v1
2. Delaigue, L., P. Landschützer, C. Wimart-Rousseau, L. A. Arbilla, J.-B. Sallée, **S. M. Bushinsky**, H. Claustre, and R. Sauzède (2026). The Southern Ocean CO₂ sink in an evolving observing system. *ESS Open Archive* 2026(0427) (). DOI: [10.22541/essoar.15002449/v1](https://doi.org/10.22541/essoar.15002449/v1)
1. Oliveli, Landschutzer, Bushinsky, submitted.

Publications

37. **Bushinsky, S. M.**, L. A. Arteaga, A. J. Fassbender, J. Hauck, M. R. Mazloff, I. Cerovecki, P. Landschützer, C. Roedenbeck, C. Danek, A. Romanou, P. Lerner, A. R. Gray, and S. Schlunegger (2026). Nonlinear Interactions of Timing and Amplitude Biases in Modeled Southern Ocean pCO₂: The Roles of Dissolved Inorganic Carbon, Total Alkalinity, and Sea Surface Temperature. *ESS Open Archive* 2026(0110) (). DOI: [10.22541/essoar.176804975.57370146/v1](https://doi.org/10.22541/essoar.176804975.57370146/v1)
36. McClish^{*}, S., **S. Bushinsky**, N. Briggs, I. Cerovečki, and C. Douglas (2026). Seasonal Re-Entrainment of Respired Organic Matter Decouples Southern Ocean Surface Production and Annual Net Community Production. en. *AGU Advances* 7(5), e2025AV002267. DOI: [10.1029/2025AV002267](https://doi.org/10.1029/2025AV002267)
35. Du, J., L. Cheng, T. Ito, H. E. Garcia, Z. Wang, J. D. Sharp, C. J. Roach, S. Minobe, Y. Zhou, B. Lu, G. G. Navarra, and **S. M. Bushinsky** (2026). Climatology and annual cycle of global ocean dissolved oxygen represented by multiple observational gridded products. English. *Ocean Science* 22(3), pp. 1483–1499. DOI: [10.5194/os-22-1483-2026](https://doi.org/10.5194/os-22-1483-2026)
34. Rustogi, P., L. Resplandy, **S. Bushinsky**, and L. Deike (2026). The influence of waves and bubbles on oxygen in the ocean interior. en. *Environmental Research Letters* 21(9), p. 094001. DOI: [10.1088/1748-9326/ae612b](https://doi.org/10.1088/1748-9326/ae612b)
33. Arteaga, L. A., C. S. Rousseaux, I. Cetinić, and **S. Bushinsky** (2026). Extreme 2016 El Niño heatwave weakened carbon export and respiration in the Equatorial Pacific. en. *Communications Earth & Environment*. DOI: [10.1038/s43247-026-03441-y](https://doi.org/10.1038/s43247-026-03441-y)

32. Ito, T., H. E. Garcia, Z. Wang, L. Cheng, J. Du, C. J. Roach, Y. Zhou, J. D. Sharp, S. K. Lauvset, S. Minobe, **S. M. Bushinsky**, B. Lu, and G. Navarra (2025). Assessing the Observational Uncertainties of Dissolved Oxygen Climatology and Seasonal Cycle Through a Coordinated Intercomparison Project. en. *Global Biogeochemical Cycles* 39(11). eprint: <https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/2025GB008751>, e2025GB008751. DOI: [10.1029/2025GB008751](https://doi.org/10.1029/2025GB008751)
31. Deike, L., X. Zhou, P. Rustogi, R. H. R. Stanley, B. G. Reichl, **S. M. Bushinsky**, and L. Resplandy (2025). A universal wind–wave–bubble formulation for air–sea gas exchange and its impact on oxygen fluxes. *Proceedings of the National Academy of Sciences* 122(38), e2419319122. DOI: [10.1073/pnas.2419319122](https://doi.org/10.1073/pnas.2419319122)
30. **Bushinsky, S. M.**, Z. Nachod*, A. J. Fassbender, V. Tamsitt, Y. Takeshita, and N. Williams (2025). Offset Between Profiling Float and Shipboard Oxygen Observations at Depth Imparts Bias on Float pH and Derived $p\text{CO}_2$. *Global Biogeochemical Cycles* 39(5), e2024GB008185. DOI: [10.1029/2024GB008185](https://doi.org/10.1029/2024GB008185)
29. Jutras #, M., **S. M. Bushinsky**, I. Cerovečki, and N. Briggs (2025). Mixing Accounts for More Than Half of Biogeochemical Changes Along Mode Water Ventilation Pathways. *Geophysical Research Letters* 52(7), e2024GL113789. DOI: [10.1029/2024GL113789](https://doi.org/10.1029/2024GL113789)
28. Dong, Y., D. C. E. Bakker, T. G. Bell, M. Yang, P. Landschützer, J. Hauck, C. Rödenbeck, V. Kitidis, **S. M. Bushinsky**, and P. S. Liss (2024). Direct observational evidence of strong CO_2 uptake in the Southern Ocean. *Science Advances* 10(30). DOI: [10.1126/sciadv.adn5781](https://doi.org/10.1126/sciadv.adn5781)
27. Ito, T., H. E. Garcia, Z. Wang, S. Minobe, M. C. Long, J. Cebrian, J. Reagan, T. Boyer, C. Paver, C. Bouchard, Y. Takano, **S. Bushinsky**, A. Cervania, and C. A. Deutsch (2024). Underestimation of multi-decadal global O_2 loss due to an optimal interpolation method. *Biogeosciences* 21(3), pp. 747–759. DOI: [10.5194/bg-21-747-2024](https://doi.org/10.5194/bg-21-747-2024)
26. Hauck, J., L. Gregor, C. Nissen, L. Patara, M. Hague, P. Mongwe, **S. Bushinsky**, S. C. Doney, N. Gruber, C. Le Quéré, M. Manizza, M. Mazloff, P. M. S. Monteiro, and J. Terhaar (2023). The Southern Ocean Carbon Cycle 1985–2018: Mean, Seasonal Cycle, Trends, and Storage. *Global Biogeochemical Cycles* 37(11), e2023GB007848. DOI: [10.1029/2023GB007848](https://doi.org/10.1029/2023GB007848)
25. Sauv  , J., A. R. Gray, C. J. Prend, **S. M. Bushinsky**, and S. C. Riser (2023). Carbon Outgassing in the Antarctic Circumpolar Current Is Supported by Ekman Transport From the Sea Ice Zone in an Observation-Based Seasonal Mixed-Layer Budget. *Journal of Geophysical Research: Oceans* 128(11), e2023JC019815. DOI: [10.1029/2023JC019815](https://doi.org/10.1029/2023JC019815)
24. Sarmiento, J. L., K. S. Johnson, L. A. Arteaga, **S. M. Bushinsky**, H. M. Cullen, A. R. Gray, R. M. Hotinski, T. L. Maurer, M. R. Mazloff, S. C. Riser, J. L. Russell, O. M. Schofield, and L. D. Talley (2023). The Southern Ocean carbon and climate observations and modeling (SOCCOM) project: A review. *Progress in Oceanography* 219, p. 103130. DOI: [10.1016/j.pocean.2023.103130](https://doi.org/10.1016/j.pocean.2023.103130)
23. McClish*, S. and **S. M. Bushinsky** (2023). Majority of Southern Ocean Seasonal Sea Ice Zone Bloom Net Community Production Precedes Total Ice Retreat. *Geophysical Research Letters*. DOI: [10.1029/2023GL103459](https://doi.org/10.1029/2023GL103459)

22. Rodgers, K. B., J. Schwinger, A. J. Fassbender, P. Landschützer, R. Yamaguchi, H. Frenzel, K. Stein, J. D. Müller, N. Goris, S. Sharma, **S. Bushinsky**, T.-T.-T. Chau, M. Gehlen, M. A. Gallego, L. Gloege, L. Gregor, N. Gruber, J. Hauck, Y. Iida, M. Ishii, L. Keppler, J.-E. Kim, S. Schlunegger, J. Tjiputra, K. Toyama, P. Vaittinada Ayar, and A. Velo (2023). Seasonal Variability of the Surface Ocean Carbon Cycle: A Synthesis. *Global Biogeochemical Cycles* 37(9), e2023GB007798. DOI: [10.1029/2023GB007798](https://doi.org/10.1029/2023GB007798)
21. Huang, Y., A. J. Fassbender, and **S. M. Bushinsky** (2023). Biogenic carbon pool production maintains the Southern Ocean carbon sink. *Proceedings of the National Academy of Sciences* 120(18). DOI: [10.1073/pnas.2217909120](https://doi.org/10.1073/pnas.2217909120)
20. Hauck, J., C. Nissen, P. Landschützer, C. Rödenbeck, **S. Bushinsky**, and A. Olsen (2023). Sparse observations induce large biases in estimates of the global ocean CO₂ sink: an ocean model subsampling experiment. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 381(2249), p. 20220063. DOI: [10.1098/rsta.2022.0063](https://doi.org/10.1098/rsta.2022.0063)
19. **Bushinsky, S. M.** and I. Cerovečki (2023). Subantarctic Mode Water Biogeochemical Formation Properties and Interannual Variability. *AGU Advances* 4(2), e2022AV000722. DOI: [10.1029/2022AV000722](https://doi.org/10.1029/2022AV000722)
Eos Research Spotlight
18. Tamsitt, V., **S. Bushinsky**, Z. Li, M. du Pleiss, S. Foppert, S. Rintoul, E. Shadwick, A. Silvano, A. Sutton, S. Swart, B. Tilbrook, and N. L. Williams (2021). Antarctica and the Southern Ocean [in “State of the Climate in 2020”]. *Bull. Amer. Meteor. Soc.* 102(8), S345–S349. DOI: [10.1175/BAMS-D-21-0081.1](https://doi.org/10.1175/BAMS-D-21-0081.1)
17. Johnson, K. S., M. F. Bif, **S. M. Bushinsky**, A. J. Fassbender, and Y. Takeshita (2020). Biogeochemical Argo [in “State of the Climate in 2019”]. *Bull. Amer. Meteor. Soc.* 101(8), S39–S41. DOI: [10.1175/BAMS-D-20-0105.1](https://doi.org/10.1175/BAMS-D-20-0105.1)
16. **Bushinsky, S. M.**, P. Landschützer, C. Rödenbeck, A. R. Gray, D. Baker, M. R. Mazloff, L. Resplandy, K. S. Johnson, and J. L. Sarmiento (2019). Reassessing Southern Ocean Air-Sea CO₂ Flux Estimates With the Addition of Biogeochemical Float Observations. *Global Biogeochemical Cycles* 33(11), pp. 1370–1388. DOI: [10.1029/2019GB006176](https://doi.org/10.1029/2019GB006176)
Eos Research Spotlight. Recognized as one of the most read papers in GBC 2018-2019
15. Arteaga, L. A., M. Pahlow, **S. M. Bushinsky**, and J. L. Sarmiento (2019). Nutrient Controls on Export Production in the Southern Ocean. *Global Biogeochemical Cycles* 33(8), pp. 942–956. DOI: [10.1029/2019GB006236](https://doi.org/10.1029/2019GB006236)
14. **Bushinsky, S. M.**, Y. Takeshita, and N. L. Williams (2019). Observing Changes in Ocean Carbonate Chemistry: Our Autonomous Future. *Current Climate Change Reports* 5(3), pp. 207–220. DOI: [10.1007/s40641-019-00129-8](https://doi.org/10.1007/s40641-019-00129-8)
13. **Bushinsky, S. M.** and S. R. Emerson (2018). Biological and physical controls on the oxygen cycle in the Kuroshio Extension from an array of profiling floats. *Deep Sea Research Part I: Oceanographic Research Papers*. DOI: [10.1016/j.dsr.2018.09.005](https://doi.org/10.1016/j.dsr.2018.09.005)

12. Gray, A. R., K. S. Johnson, **S. M. Bushinsky**, S. C. Riser, J. L. Russell, L. D. Talley, R. Wanninkhof, N. L. Williams, and J. L. Sarmiento (2018). Autonomous Biogeochemical Floats Detect Significant Carbon Dioxide Outgassing in the High-Latitude Southern Ocean. *Geophysical Research Letters* 45(17), pp. 9049–9057. DOI: [10.1029/2018GL078013](https://doi.org/10.1029/2018GL078013)

Recognized as one of the most read papers in GRL 2018-2019

11. **Bushinsky, S. M.**, A. R. Gray, K. S. Johnson, and J. L. Sarmiento (2017). Oxygen in the Southern Ocean From Argo Floats: Determination of Processes Driving Air-Sea Fluxes. *Journal of Geophysical Research: Oceans* 122(11), pp. 8661–8682. DOI: [10.1002/2017JC012923](https://doi.org/10.1002/2017JC012923)

Editor's Highlight

10. Yang, B., S. R. Emerson, and **S. M. Bushinsky** (2017). Annual net community production in the subtropical Pacific Ocean from in situ oxygen measurements on profiling floats. *Global Biogeochemical Cycles* 31(4), pp. 728–744. DOI: [10.1002/2016GB005545](https://doi.org/10.1002/2016GB005545)
9. Newsom, E. R., A. J. Fassbender, A. E. Maloney, and **S. M. Bushinsky** (2016). Increasing the usability of climate science in political decision-making. *Elementa: Science of the Anthropocene* 4, p. 000127. DOI: [10.12952/journal.elementa.000127](https://doi.org/10.12952/journal.elementa.000127)
8. Emerson, S. and **S. Bushinsky** (2016). The role of bubbles during air-sea gas exchange. *Journal of Geophysical Research: Oceans* 121(6), pp. 4360–4376. DOI: [10.1002/2016JC011744](https://doi.org/10.1002/2016JC011744)

7. **Bushinsky, S. M.**, S. R. Emerson, S. C. Riser, and D. D. Swift (2016). Accurate oxygen measurements on modified Argo floats using in situ air calibrations. *Limnology and Oceanography: Methods* 14(8), pp. 491–505. DOI: [10.1002/lom3.10107](https://doi.org/10.1002/lom3.10107)

Editor's Highlight

6. **Bushinsky, S. M.** and S. Emerson (2015). Marine biological production from in situ oxygen measurements on a profiling float in the subarctic Pacific Ocean. *Global Biogeochemical Cycles* 29(12), pp. 2050–2060. DOI: [10.1002/2015GB005251](https://doi.org/10.1002/2015GB005251)

Cover Article

5. Emerson, S. and **S. Bushinsky** (2014). Oxygen Concentrations and Biological Fluxes in the Open Ocean. *Oceanography* 27(1), pp. 168–171. DOI: [10.5670/oceanog.2014.20](https://doi.org/10.5670/oceanog.2014.20)
4. **Bushinsky, S. M.** and S. Emerson (2013). A method for in-situ calibration of Aanderaa oxygen sensors on surface moorings. *Marine Chemistry* 155, pp. 22–28. DOI: [10.1016/j.marchem.2013.05.001](https://doi.org/10.1016/j.marchem.2013.05.001)
3. Howard, E., S. Emerson, **S. Bushinsky**, and C. Stump (2010). The role of net community production in air-sea carbon fluxes at the North Pacific subarctic-subtropical boundary region. *Limnology and Oceanography* 55(6), pp. 2585–2596. DOI: [10.4319/lo.2010.55.6.2585](https://doi.org/10.4319/lo.2010.55.6.2585)
2. Arrigo, K. R., G. L. van Dijken, and **S. Bushinsky** (2008). Primary production in the Southern Ocean, 1997-2006. *Journal of Geophysical Research: Oceans* 113(C08004). DOI: [10.1029/2007JC004551](https://doi.org/10.1029/2007JC004551)

1. Micheli, F., A. O. Shelton, **S. M. Bushinsky**, A. L. Chiu, A. J. Haupt, K. W. Heiman, C. V. Kappel, M. C. Lynch, R. G. Martone, R. B. Dunbar, and J. Watanabe (2008). Persistence of depleted abalones in marine reserves of central California. *Biological Conservation* 141(4), pp. 1078–1090. DOI: [10.1016/j.biocon.2008.01.014](https://doi.org/10.1016/j.biocon.2008.01.014)

White Papers

2. **Bushinsky, SM**, RC Hamme, DP Nicholson, KS Johnson (2017). Oxygen measurements from autonomous vehicles: applications and challenges. *Autonomous and Lagrangian Platforms and Sensors II*.
1. Wanninkhof, R, K Johnson, N Williams, J Sarmiento, S Riser, E Briggs, **S Bushinsky**, B Carter, A Dickson, R Feely, A Gray, L Juranek, R Key, L Talley, J Russel, A Verdy (2016). An evaluation of pH and NO₃ sensor data from SOCCOM floats and their utilization to develop ocean inorganic carbon products. Southern Ocean Carbon and Climate Observations and Modeling Carbon Working Group.

Datasets

1. Landschützer, Peter; **Bushinsky, Seth**; Gray, Alison R. (2019). A combined globally mapped CO₂ flux estimate based on the Surface Ocean CO₂ Atlas Database (SOCAT) and Southern Ocean Carbon and Climate Observations and Modeling (SOCCOM) biogeochemistry floats from 1982 to 2017 (NCEI Accession 0191304). Version 2.2. NOAA National Centers for Environmental Information. Dataset. doi:10.25921/9hsn-xq82 [2019-07-17].

Invited presentations since July 2019

- 2026** Princeton University Climate Seminar. Princeton, NJ, Sept. 2026 (**Talk**).
 Geophysical Fluid Dynamics Laboratory. Princeton, NJ. Sept. 2026 (**Talk**)
 Alfred Wegener Institute - Institute for Polar Science. Bremerhaven, Germany. June 2026. (**Talk**).
 Swiss Ocean Day. Bern, Switzerland. Scheduled for June 2026 (**Keynote Talk**).
 Vlaams Instituut voor de Zee (VLIZ). Ostend, Belgium. May 2026 (**Talk**)
 European Geophysical Union. Vienna, Austria. May 2026 (**Talk**)
 Eawag: Swiss Federal Institute of Aquatic Science and Technology, Kastanienbaum.
 Surface Waters Seminars. Lucerne, Switzerland. April 2026 (**Talk**).
 University of Bern. Bern, Switzerland. April 2026 (**Talk**).
 University of Basel, Department of Environmental Sciences Seminar. Basel, Switzerland. March 2026 (**Talk**).
- 2025** Mid-latitude Atmosphere-Ocean-Ecosystem Interactions: Processes, Predictability, and Habitability, July 2025 (**Talk**).
 Global Ocean Oxygen Network (IOC Expert Working Group GO2NE) webinar on ocean deoxygenation, May 2025 (**Talk**).
 El Nino-Southern Oscillation: Past, Present, and Future; Celebrating the Scientific Legacy of Klaus Wyrtki, March 2025 (**Keynote Talk**).
- 2024** 20th Chinese-American Kavli Frontiers of Science Symposium, November 2024 (**Poster**).
 Xiamen University Department of Ocean Sciences Seminar, November 2024 (**Talk**).
 Global Ocean Biogeochemistry Array Annual Meeting, September 2024 (**Talk**).

- Southern Ocean Carbon and Climate Observations and Modeling project 10th anniversary meeting, June 2024 (**Talk**)
- 2023** Gordon Research Conference in Chemical Oceanography, July 2023 (**Talk**).
Southern Ocean Carbon and Climate Observations and Modeling project Annual Meeting, Princeton University, June 2023 (**Talk**).
NASA Joint Science Workshop, College Park, MD, May 2023 – slides shared, unable to attend due to COVID (**Talk**).
- 2022** Ocean Carbon and Biogeochemistry Summer Workshop, June 2022 (**Talk**).
NOAA Biogeochemical Argo / Fisheries Workshop, June 2022 (**Talk**).
- 2021** Global Ocean Biogeochemistry Array Annual Meeting, September 2021 (**Talk**).
Southern Ocean Carbon and Climate Observations and Modeling project annual meeting, June 2021 (**Talk**).
Global Ocean Biogeochemistry Array Workshop, “Building a Community of Biogeochemical Float Data Users”, June 2021 (**Talk**).
Ocean Carbon and Biogeochemistry Carbon Gaps Working Group Year 2 Meeting, May 2021 (**Talk**).
Old Dominion University, Department of Ocean & Earth Sciences Seminar, April 2021 (**Talk**).
Global Carbon Budget Ocean Workshop, March 2021 (**Talk**).
- 2020** Southern Ocean Carbon and Climate Observations and Modeling project annual meeting, June 2020 (**Talk**).
Ocean Sciences Meeting 2020 Session: Seasonal Cycles of Ocean Biogeochemistry and Ecosystems Under a Changing Climate (**Poster**).
- 2019** The 3rd Joint Institute of Marine and Atmospheric Research Symposium, “New Technologies Driving Ocean Science Breakthroughs”, November 2019 (**Talk**).

RESEARCH SUPPORT

Current External Funding

- Title: SOCCOM3: Science to predict the future of the Southern Ocean and its global impacts
- Role: **Co-PI/Institutional PI**
- Agency: National Science Foundation: Polar Programs
- Collaborators: Lead PI L. Talley, S. Purkey (Scripps Institution of Oceanography). J. Russell (University of Arizona), K. Johnson, Y. Takeshita, (Monterey Bay Aquarium Research Institute), O. Schofield (Rutgers University), S. Riser, A. Gray (University of Washington)
- Title: Integration of models and observations across scales (InMOS): A novel data synthesis framework for diagnosing the coupled cycles of ocean carbon, oxygen, and heat
- Role: **Co-PI / Observational Lead**
- Agency: Schmidt Sciences: Ocean Biogeochemistry Virtual Institute

Collaborators: Lead PI / State Estimate Lead: T. DeVries (UC Santa Barbara), co-Lead PI: R. Keeling (Scripps Institution of Oceanography), Modeling Lead: L. Resplandy (Princeton University), L. Zanna (NYU), P. Landschützer (Flanders Marine Institute), P. Hassanzadeh (U. Chicago), C. Rödenbeck (Max Planck Institute-Jena), C. Labuschagne (South African Weather Service), F. Primeau (UC Irvine), H. Graven (Imperial College).

Title: Advancing satellite-constrained modeled air-sea CO₂ fluxes with a focus on the strength of the Southern Ocean carbon sink

Role: **Co-I/Institutional PI**

Agency: National Aeronautics and Space Administration: Carbon Monitoring System

Collaborators: Lead PI: L. Arteaga (NASA/ University of Maryland Baltimore County), Co-I C. Rousseaux (NASA Goddard), Collaborator L. Ott (NASA Goddard)

Title: Collaborative Research: Physical and biological controls on ocean carbon and oxygen uptake in the Western North Pacific

Role: **Lead PI**

Agency: National Science Foundation: Chemical Oceanography, Physical Oceanography, and EPSCOR

Duration: September 2021 – August 2024, 1 year NCE through 2025

Collaborators: co-PIs: M. Manizza, I. Cerovečki (Scripps Institution of Oceanography)

Title: Quantifying the role of mode and intermediate waters in Southern Ocean carbon fluxes through the use of profiling floats, satellite observations, and a state estimate

Role: **Lead PI**

Agency: National Aeronautics and Space Administration: Carbon Cycle Science

Collaborators: co-PI: I. Cerovečki (Scripps Institution of Oceanography)

Past External Funding

Title: Biogeochemical Argo synthesis products of oxygen, nitrate, and pH for increased community utilization of autonomous profiling observations

Role: **Lead PI**

Agency: National Oceanographic and Atmospheric Administration

Collaborators: co-PIs: N. Williams (University of South Florida), A. Fassbender (NOAA-PMEL).

Title: Characterizing the impact of seasonal sea ice on phytoplankton blooms and net community production with integrated satellite and biogeochemical profiling float observations

Role: **PI for FINESST Graduate Student Fellowship for Future Investigator Shannon McClish (advisee)**

Agency: National Aeronautics and Space Administration

Title: Determination of the Relationship between primary Production and Net Community Production in Southern Ocean through the use of profiling floats, satellite data, and ecosystem models

Role: **Co-I/Institutional PI**

Agency: National Aeronautics and Space Administration

Collaborators: Lead-PI: Jorge Sarmiento (Princeton University), co-I/Institutional PI: L. Arteaga (NASA/ University of Maryland Baltimore County).

Internal UHM Funding

Title: Visualizing the 3-dimensional structure of ocean pathways and biogeochemical changes

Role: **Co-Mentor with postdoc Mathilde Jutras**

Agency: Summer Undergraduate Research and Creative Work Mentoring Grant

Year: 2024

TEACHING

Graduate Courses Taught

OCN 623: Chemical Oceanography: 2020, 2022, 2023, 2024, 2025

OCN 780: Oceanography Seminar: 2020

Undergraduate Courses Taught

OCN 201: Science of the Sea: 2021, 2022, 2023, 2024, 2025

OCN 201L: Laboratory Course paired with OCN 201: 2021, 2022, 2023

MENTORING

UHM Advisees

Current

Graduate students, committee chair

3. Ally Morris: 2025-, PhD
2. Charles Addey: 2022-, PhD
1. Zachary Nachod: 2022-, PhD

Postdoctoral scholars

2. Raphaël Bajon: Nov. 2025-
1. Haichao Guo: July 2025-

Undergraduate students

1. Jenny Joy Mactumbas (UROP funding recipient): 2023-

Past

Graduate students, committee chair or co-chair

2. Shannon McClish: 2020-2025, PhD.
1. James Ash, MS: 2021-2023. MS Committee Co-Chair w/ Angelique White

Postdoctoral scholars

2. Daniela König: April 2023-December 2025. Now a postdoctoral researcher with Prof. Nicholas Hawco, UH Mānoa.

1. Mathilde Jutras: September 2023-May 2025. Funded in part by Canadian government fellowship. Now an Assistant Professor at Remouski, UQCAR, Canada.

Undergraduate students

3. Tarja Luna Ogando (Thesis advisor), Defended Spring 2025
2. Kamryn Shimizu, 2022
1. Anthony Barro (Thesis advisor): Defended Fall 2020

External Advisees

Current

1. Marc Pyolle (UNIVERSITÉ DU QUÉBEC À RIMOUSKI, Institut des sciences de la mer (ISMER), Canada): 2023-2024, PhD Committee member and thesis examiner.

Past

2. Simin Kheradmand (University of Victoria, BC, Canada): 2023-2024, PhD Committee member
1. Kacy Kim (Hawai'i Pacific University). 2022. MS Committee member

SERVICE

Departmental

OCN Faculty Search Committees

Sept. 2022 – Aug. 2023	Uehiro Faculty Search Committee member
Jul. 2022 – Jan. 2023	Marine Geology and Geochemistry Faculty Search Committee member
Jan. – Mar. 2020	Mega-search (four position) Committee
Dec. 2019	Deep Sea Ecologist Search Committee

OCN Committees

July 2019 -	Non-Advisee: Comprehensive Exam Committees, (x11); PhD Committees (x3); MS Committees (x3); Advisory Committees (x5)
Feb. 2021 –	Kahuliau / Interim Diversity Equality and Inclusion Committee member, Fall 2022 SEED IDEAs application for DEI speaker honoraria, Coordination of OCN Seminar DEI speakers, Facilitated vote on increasing grad. student pay and indexing to cost of living
May 2020 –	Teaching Evaluation Committee, member
Fall 2022	TA Assignment Committee

Marine Geology & Geochemistry (MGG) Division

Fall 2023 –	Biogeochemistry Brown Bag (semi-weekly informal seminar centered on biogeochemistry within the division/department) faculty organizer
Winter 2022	MGG Division application review graduate student selection committee
Winter 2021	MGG Division application review graduate student selection committee

Feb. 2021 – May 2021 MGG Admissions Guideline Creation Committee

General Departmental Service

August 2021	SOEST Speed Networking faculty participant
Jun. 2020 – Sept. 2021	Provided the “Graduate Student Agreement” I developed for my group to the students drafting the new Oceanography “Student-Advisor Contract”. Gave feedback on an early draft of the developing document
Jan. 2021 – May 2022	Unlearning Racism in Geosciences (URGE) Pod Leader
2020, 2021	Research Presentations to OCN 100: Global Environmental Science Seminar
Feb. 2020	Search advocate training
2019-present	GES undergraduate symposium evaluator (3 students total)
July 2019	Presentation to Mauka to Makai Bridge summer course entitled “Understanding ocean biogeochemistry using robots.”

SOEST / University

Sept. 2023 - July 2024	ORE Faculty Search Committee member
Fall 2023, Fall 2024	SOEST/UHM Open House exhibitions, Lego profiling float competition

National/International

2026	National Academy of Sciences working group committee member. The future of Biogeochemical Argo.
2022	Ocean Sciences Meeting Co-Chair of HL06 Advances in understanding the circulation and carbon cycle of the Southern Ocean
2020-2022	Ocean Carbon and Biogeochemistry Scientific Steering Committee early career member, 2020-2021; regular member, 2022.
Sept. 2021	Adviser for ocean carbon cycle portion of National Public Radio news story: Your Next Car May Be Built With Ocean Rocks. Scientists Can’t Agree If That’s Good. Contacted by Neela Banerjee, Supervising Climate Editor.

Journal peer reviewer (n=53) for:

Biogeosciences	Geophysical Research Letters (x7)
Deep Sea Research Part I (x2)	Elementa
Frontiers in Marine Science	ESSD
Global Biogeochemical Cycles (x9)	Internal NOAA reviews (x2)
Geosciences	Journal of Sea Research

Journal of Atmospheric and Oceanic Technology	Nature Reviews Earth and Environment
Journal of Geophysical Research (JGR): Atmosphere	Ocean Science
JGR: Oceans (x8)	Ocean Science Discussions
Limnology and Oceanography: Methods (x5)	Phil. Trans. Royal Soc. A
Methods in Oceanography	Polar Research
Nature Communications (x2)	Progress in Oceanography
Nature Geosciences	Science

Proposal reviewer (n=21) for:

National Science Foundation (Chemical Oceanography (x6), Polar Programs, Ocean technology (x3), Physical Oceanography (x3), Postdoc Res. Fellowship, CAREER)	tration (Climate Observations and Monitoring, Ocean Acidification Program (x3))
National Ocean Partnership Program	Dutch Research Council
National Oceanic and Atmospheric Administration	Schmidt Sciences

External thesis/promotion reviewer for:

Scripps Institution of Oceanography, USA
University of Cape Town, South Africa

MEDIA/NEWS ABOUT MY RESEARCH

June 2024	The Current: UC Santa Barbara leads \$9 million research project on ocean cycles, https://news.ucsb.edu/2024/021495/uc-santa-barbara-leads-9-million-research-project-ocean-cycles
June 2024	SOEST News: \$4.5M awarded to UH to advance understanding of ocean carbon, oxygen, heat, https://www.soest.hawaii.edu/soestwp/announce/news/4-5m-schmidt-obvi/
June 2024	UH News: \$4.5M to advance understanding of ocean carbon, oxygen, heat, https://www.hawaii.edu/news/2024/06/14/4-5m-ocean-carbon-oxygen-heat/
April 2023	NOAA Research: One of The Planet's Most Important Carbon Sinks is Revealing its Secrets, https://research.noaa.gov/2023/04/26/one-of-the-planets-most-important-carbon-sinks-is-revealing-its-secrets/
March 2023	Eos: A deeper dive into wintry, carbon-absorbing Antarctic waters, https://doi.org/10.1029/2023EO230096
October 2021	NOAA Climate Program Office notice of funding, https://cpo.noaa.gov/Funding-Opportunities/FY2021-Recipients
October 2021	UH News: \$2.7M for ocean floats, satellites to give UH scientists new ocean, climate insights. https://www.hawaii.edu/news/2021/10/15/ocean-floats-satellites-grants/
November 2019	Eos: Ship-Based Measurements Overestimate Southern Ocean Carbon Sink. https://eos.org/research-spotlights/ship-based-measurements-overestimate-southern-ocean-carbon-sink

- June 2019 OCB Science Highlights: A new era of observing the ocean carbonate system. <https://www.us-ocb.org/a-new-era-of-observing-the-ocean-carbonate-system/>
- June 2019 Science News: The Southern Ocean may be less of a carbon sink than we thought. <https://www.sciencenews.org/article/southern-ocean-antarctica-absorbs-less-carbon-expected?tgt=nr>
- May 2019 Chemical & Engineering News: How robots are revolutionizing chemical oceanography. <https://cen.acs.org/environment/water/Podcast-robots-revolutionizing-chemical-oceanography/97/i21>
- December 2018 Nature News: Massive ocean carbon sink spotted burping CO₂ on the sly, <https://www-nature-com.eres.library.manoa.hawaii.edu/articles/d41586-018-07784-1>

OCEANOGRAPHIC CRUISES

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|------|---------|------------------------|--|
| 2013 | 3 days | R/V Thomas G. Thompson | Deployment of mooring-based oxygen calibration system on Che'ba mooring. |
| 2013 | 19 days | CCGS John P. Tully | Calibration, deployment, and recovery of oxygen, nitrogen, and pH sensors at Ocean Station Papa. |
| 2013 | 21 days | R/V Melville | Calibration and deployment of 10 SOS-Argo floats in the Kuroshio Extension. |
| 2012 | 19 days | CCGS John P. Tully | Calibration, deployment, and recovery of oxygen, nitrogen, and pH sensors at Ocean Station Papa. Deployment of initial test SOS-Argo float. |
| 2011 | 7 days | USNS Safeguard | Calibration, deployment, and recovery of oxygen, nitrogen, and pH sensors at Kuroshio Extension Observatory. |
| 2011 | 19 days | CCGS John P. Tully | Calibration, deployment, and recovery of oxygen, nitrogen, and pH sensors at Ocean Station Papa. |
| 2010 | 19 days | CCGS John P. Tully | Calibration, deployment, and recovery of oxygen, nitrogen, and pH sensors at Ocean Station Papa. |
| 2009 | 19 days | CCGS John P. Tully | Calibration, deployment, and recovery of oxygen, nitrogen, and pH sensors at Ocean Station Papa. Deployment and recovery of Sea Gliders with oxygen sensors. |
| 2008 | 23 days | R/V Thomas G. Thompson | Synthesis and analysis of underway data, sample collection. |